⌈Thm. Let $f:[a,b]$ be a monotone increasing function.
Then, for any $x\in(a,b)$, $f(x-)$ and $f(x+)$ exist with

$$\sup_{a<t<x} f(t) = f(x-) \le f(x) \le f(x+) \le \inf_{x<t<b} f(t)$$

Moreover, if $a\le x<y\le b$, then $f(x+)\le f(y-)$.

Proof. Since f is mono. increasing, for any $t\in(a,x)$, $f(t)\le f(x)$.
Therefore, $\sup_{a<t<x} f(t) = \sup\{f(t) \mid a<t<x\} \le f(x)$.

Given $\varepsilon>0$, there exists $\delta>0$ such that

$$\sup_{a<t<x} f(t) - \varepsilon < f(x-\delta) \le \sup_{a<t<x} f(t)$$

Now, for any $s\in(x-\delta, x)$, $f(x-\delta)\le f(s)$, therefore

$$\sup_{a<t<x} f(t) - f(s) < \varepsilon$$

so, $f(x-) = \sup_{a<t<x} f(t)$. ⌋

Meanwhile, the second assertion can be proved by:

$$f(x^+) = \inf_{x<t<b} f(t) = \inf_{x<t<y} f(t) \quad\text{and}\quad f(y^-) = \sup_{a<t<y} f(t) = \sup_{x<t<y} f(t)$$

$\boxtimes$

⌈Corollary. A discontinuous of monotone function is at most countable.
Proof. Let $f:(a,b)\to\mathbb{R}$ be a monotone increasing function, E be the set of discontinuity points.
Since f is monotone, $x\in E$ if and only if $f(x-)<f(x+)$. By density of rationals, given x,

$$f(x-) < r(x) < f(x+)$$

for some rational $r(x)\in\mathbb{Q}$. And if $x_1<x_2$, then

$$r(x_1) < f(x_1+) \le f(x_2-) < r(x_2)$$

implies $r(x_1)<r(x_2)$. Therefore, there is injective map from E into $\mathbb{Q}$.

「Thm. If $f:(a,b)\to\mathbb{R}$ satisfies Intermidate Value Property, then f cannot have first-kind discontinuity.

Proof. Suppose that f has first-kind discontinuity at $x\in(a,b)$.
Then, $f(x^-)$ and $f(x^+)$ exist, but $f(x^-)=f(x)=f(x^+)$ cannot be happen.
WLOG, suppose $f(x^-)\neq f(x)$. Put $\epsilon=|f(x^-)-f(x)|>0$.
Since $f(x^-)$ exists, there exists $\delta>0$ s.t $t\in(x-\delta,x)\Rightarrow|f(x^-)-f(t)|<\frac{\epsilon}{3}$.
Now, if $f(x)>f(x^-)$, then

$$f(x-\tfrac{\delta}{2})<f(x-\tfrac{\delta}{2})+\tfrac{2}{3}\epsilon<f(x)$$

But there is no $c\in(x-\frac{\delta}{2},x)$ s.t $f(c)=f(x-\frac{\delta}{2})+\frac{2}{3}\epsilon$, because $|f(c)-f(x-\frac{\delta}{2})|<\frac{\epsilon}{3}$. If $f(x)<f(x^-)$, similarly contradiction.